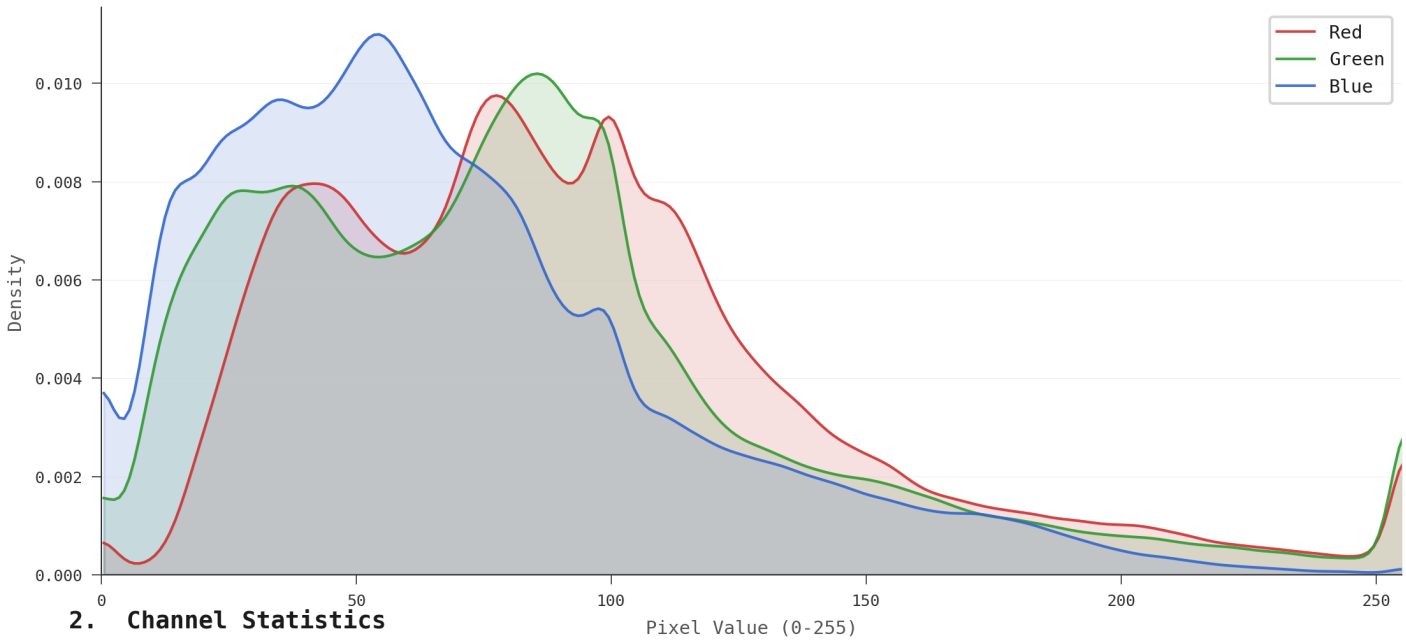
Reference Data Sheet: KODIM05

Subject: Motocross Starting Lineup

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	91.5034	49.0693	2407.8009	0.9401	0.9473	0	255
Green	81.0216	51.4019	2642.1585	1.2491	1.0591	0	255
Blue	67.5160	44.7156	1999.4804	0.9117	1.0294	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8981	0.8215
G	0.8981	1.0000	0.9034
B	0.8215	0.9034	1.0000

Pairwise Pearson Correlation Coefficients

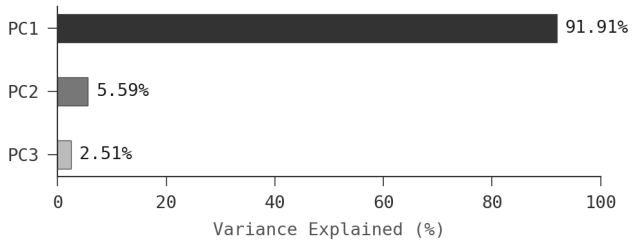
r(R-G)	=	**0.898130**
r(R-B)	=	**0.821497**
r(G-B)	=	**0.903419**
Avg |r|	=	**0.874349**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{2407.8071} & 2265.3215 & 1802.5027 \\
2265.3215 & \mathbf{2642.1652} & 2076.4827 \\
1802.5027 & 2076.4827 & \mathbf{1999.4855}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM05

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5794	**-0.6251**	-0.5230	6478.8129
PC2	-0.7287	0.1100	0.6759	394.0318
PC3	0.3651	-0.7727	0.5193	176.6130

6. Derived Metrics

Condition Number

36.68

(l1/l3) PC1/PC2: 16.44 | PC2/PC3: 2.23

Blue Channel Independence

11.4%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6251 | Green dominant

Classification: Weakly One-Dimensional (PC1 = 91.91%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.904243	0.892715	0.898479
Green	0.908033	0.895850	0.901942
Blue	0.891080	0.877531	0.884306

8. Average Local Variance (3x3)

Channel	Value
Red	389.2025
Green	420.3867
Blue	360.1707

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 36.68 | PC1: 91.91%

Correlation: G–B pair leads at r = 0.903419 | R–G: 0.898130 | R–B: 0.821497

Blue Channel: 11.4% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.894909 | Avg local variance: 389.9200

Eigenvector: Green dominant | G (-0.6251) leads | R (-0.5794), B (-0.5230)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."